

Regularizations

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Slide Credits to Shusen Wang

The ℓ_2 -Norm Regularization

Linear Regression

Input: feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and labels $\mathbf{y} \in \mathbb{R}^n$.

Output: vector $\mathbf{w} \in \mathbb{R}^d$ such that $\mathbf{X}\mathbf{w} \approx \mathbf{y}$.

Task

Linear Regression

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Task

- Least squares regression:

$$\min_{\mathbf{w}} \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2.$$

- Ridge regression:

$$\min_{\mathbf{w}} \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \gamma \|\mathbf{w}\|_2^2.$$



Loss Function

Regularization

Methods

Ridge Regression:

Algorithms

- **Analytical solution:** $\mathbf{w}^\star = (\mathbf{X}^T \mathbf{X} + n\gamma \mathbf{I}_d)^{-1} \mathbf{X}^T \mathbf{y}$.
 - Time complexity: $O(nd^2 + d^3)$.

Ridge Regression:

Algorithms

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 - Time complexity: $O(nd^2 + d^3)$.
- **Derivations:**
 - The objective function is $Q(\mathbf{w}) = \frac{1}{n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \gamma \|\mathbf{w}\|_2^2$.
 - The gradient is $\nabla Q(\mathbf{w}) = \frac{2}{n} \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + 2\gamma \mathbf{w}$.
 - Set $\nabla Q(\mathbf{w}^*) = 0$ leads to $\frac{2}{n} (\mathbf{X}^T \mathbf{X} + n\gamma \mathbf{I}_d) \mathbf{w}^* = \frac{2}{n} \mathbf{X}^T \mathbf{y}$.
- **Time complexity:**
 - $O(nd^2)$ time for the multiplication $\mathbf{X}^T \mathbf{X}$.
 - $O(d^3)$ time for the inversion of the $d \times d$ matrix $\mathbf{X}^T \mathbf{X} + n\gamma \mathbf{I}_d$.

Ridge Regression:

Algorithms

- **Conjugate gradient (CG)**

- $O\left(\sqrt{\kappa} \log \frac{n}{\epsilon}\right)$ iterations to reach ϵ precision.
- Hessian matrix: $\nabla^2 Q(\mathbf{w}) = \frac{2}{n}(\mathbf{X}^T \mathbf{X} + n\gamma \mathbf{I}_d)$.
- $\kappa = \frac{\lambda_{\max}(\mathbf{X}^T \mathbf{X}) + n\gamma}{\lambda_{\min}(\mathbf{X}^T \mathbf{X}) + n\gamma}$ is the condition number of the Hessian.

Usefulness of Regularization

Question: Why do we use the ℓ_2 -norm regularization?

Usefulness of Regularization

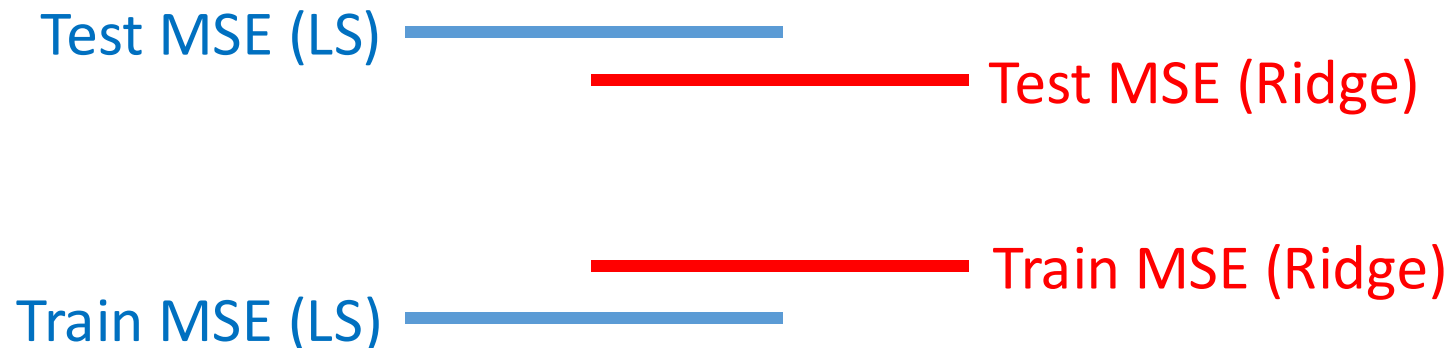
Question: Why do we use the ℓ_2 -norm regularization?

- Reason 1: easier to optimize.
 - Conjugate gradient (CG) requires $O\left(\sqrt{\kappa} \log \frac{n}{\epsilon}\right)$ iterations to reach ϵ precision.
 - Least squares: $\kappa = \frac{\lambda_{\max}(\mathbf{X}^T \mathbf{X})}{\lambda_{\min}(\mathbf{X}^T \mathbf{X})}$.
 - Ridge regression: $\kappa = \frac{\lambda_{\max}(\mathbf{X}^T \mathbf{X}) + n\gamma}{\lambda_{\min}(\mathbf{X}^T \mathbf{X}) + n\gamma}$. ($\gamma \uparrow$, $\kappa \downarrow$).
 - ➡ CG converges faster as γ increases.

Usefulness of Regularization

Question: Why do we use the ℓ_2 -norm regularization?

- Reason 1: easier to optimize.
- Reason 2: better generalization.
 - Least squares has better training error (due to the optimality).
 - Ridge regression makes better prediction on test set (due to *bias-variance decomposition*).



The ℓ_1 -Norm Regularization

Motivations

$$\mathbf{x} \in \mathbb{R}^d \xrightarrow{\text{prediction}} y \in \mathbb{R}$$

Fact 1: y can be independent of some of the d feature.

Fact 2: if $d \gg n$, linear models are likely to overfit.

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Example: Use genomic data to predict disease.

- d is huge: human have 20K protein-coding genes.
- n is small: tens or hundreds of human participants in an experiment.
- Most genes are irrelevant to a specific disease.

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Goal 1: Select the features relevant to y .

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Fact 2: if $d \gg n$, linear models are likely to overfit.

Goal 1: Select the features relevant to y .

Goal 2: Prevent overfitting for *large d , small n* problems.

The ℓ_1 -Norm Constraint

• LASSO: $\min_{\mathbf{w}} \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2;$ s. t. $\|\mathbf{w}\|_1 \leq t.$

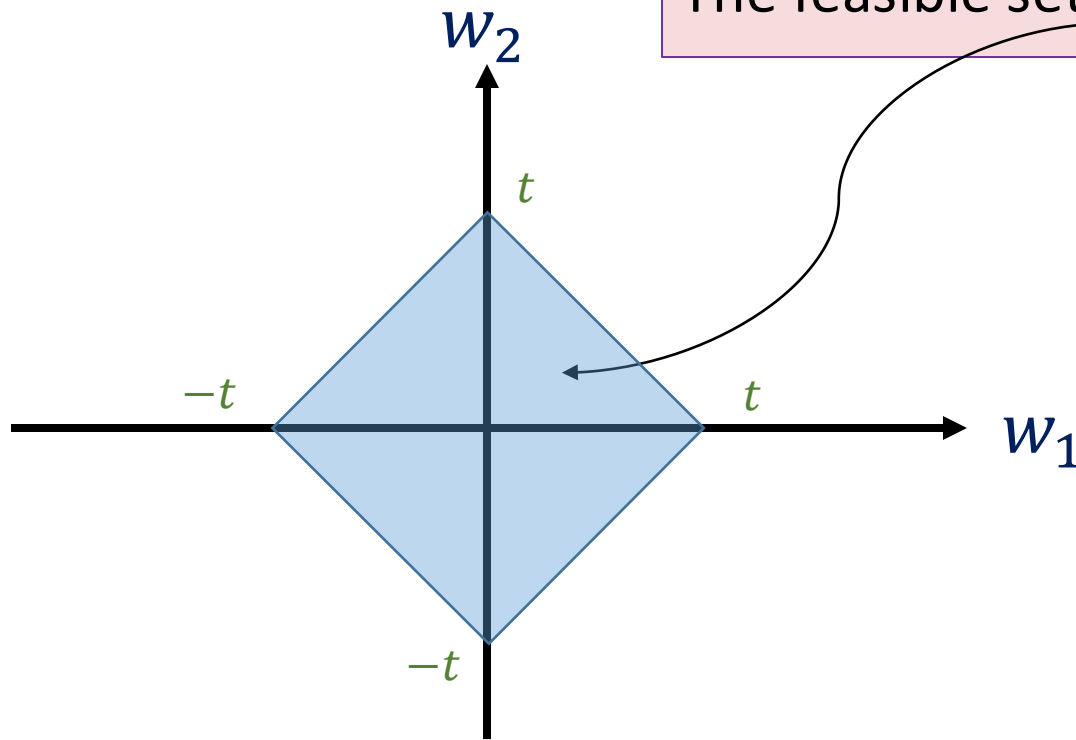


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 - Smaller $t \rightarrow$ sparser $\mathbf{w}^*$.

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 - Smaller $t \rightarrow$ sparser $\mathbf{w}^*$.
 - Sparsity $\longleftrightarrow$ feature selection. Why?
 - Let $\mathbf{x}'$ be a test feature vector.
 - The prediction is $\mathbf{x}'^T \mathbf{w}^* = w_1^* x'_1 + w_2^* x'_2 + \dots + w_d^* x'_d.$
 - If $w_1^* = 0$, then the prediction is independent of x'_1 .

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- Another form: $\min_{\mathbf{w}} \frac{1}{2n} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \gamma \|\mathbf{w}\|_1.$



Loss Function



Regularization

Summary

Regularized ERM

- Regularized empirical risk minimization:

$$\min_{\mathbf{w} \in \mathbb{R}^d} \quad \frac{1}{n} \sum_{i=1}^n L(\mathbf{w}; \mathbf{x}_i, y_i) \quad + \quad R(\mathbf{w}).$$

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Loss Function

- Linear regression: $L(\mathbf{w}; \mathbf{x}_i, y_i) = \frac{1}{2} (\mathbf{w}^T \mathbf{x}_i - y_i)^2$
- Logistic regression: $L(\mathbf{w}; \mathbf{x}_i, y_i) = \log(1 + \exp(-y_i \mathbf{w}^T \mathbf{x}_i))$
- SVM: $L(\mathbf{w}; \mathbf{x}_i, y_i) = \max\{0, 1 - y_i \mathbf{w}^T \mathbf{x}_i\}$

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Regularization

- ℓ_1 -norm: $R(\mathbf{w}) = \gamma ||\mathbf{w}||_1$
- ℓ_2 -norm: $R(\mathbf{w}) = \gamma ||\mathbf{w}||_2^2$
- Elastic net: $R(\mathbf{w}) = \gamma_1 ||\mathbf{w}||_1 + \gamma_2 ||\mathbf{w}||_2^2$